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Stimulating Consumption at Low Budget: Evidence from a Large-Scale Policy Experiment Amid the COVID-19 Pandemic

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Abstract. We use a novel panel with detailed transaction records of more than one million de-identified individuals to study the effect of a large-scale Chinese government-issued digital coupon program on consumer spending. Exploiting a difference-in-differences approach, we find that the digital coupon is highly effective in stimulating consumption. An effective government subsidy of RMB 1 can drive excess spending of RMB 3.4 to RMB 5.8, and the effect persists across multiple coupon issuance waves. In explaining the results, we find that a behavioral model with mental accounting and loss aversion can match the empirical evidence from the field. Our analysis, by illustrating the importance of embedding behavioral factors into the design and implementation of public policy, informs the current debate about cost-effective policy tools to recover the economy.

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Keywords: fiscal stimulus program • digital coupon • behavioral economics • policy experiment • MPC

1. Introduction

How to spur consumer spending and stimulate the economy is a critical question facing governments in economic downturns. Oftentimes governments resort to stimulus payments, such as tax rebate or cash payment, as a response. In this paper, we study an innovative policy tool widely adopted by the local governments in China to stimulate consumption in the wake of the COVID-19 pandemic: the digital coupon program. Whereas coupon as a price discrimination and advertising vehicle to attract new consumers and increase purchases has been widely adopted by merchants and is well-studied (see, e.g., Narasimhan 1984, Varian 1989, Reimers and Xie 2019), government-issued shopping coupon as a consumption stimulus is not common, and its effectiveness is largely unknown.¹

Since mid-March 2020, more than 200 cities in 31 provinces of China have rolled out their digital coupon programs (as of June 2020) in an attempt to revive local retail, which has been severely affected by the COVID-19 pandemic. This coupon program departs from other commonly adopted fiscal stimulus programs such as cash payment in several salient ways. First, the coupon typically takes the form of saving with certain amount of spending, for example, “spend RMB 40, get RMB 10

off”, and hence, has the nature of “use it or lose it”. Second, unlike the previous government-initiated coupon programs that involve relatively large amounts in hundreds of dollars and redemption period of several months (Hsieh et al. 2010, Kan et al. 2017), the coupons in this context are of small face value and short duration in the time frame of one to two weeks and involve a small amount of government subsidy. Third, the coupons are disbursed in waves through mobile payment platforms with limited quantity in each round on a “first-come-first-serve” basis.

The experiments of the digital coupon by local governments in an attempt to boost consumption offer a rich ground for research. Understanding how consumers respond to such small value, short-duration digital coupons and the underlying mechanisms is important for both academic research and policy-making. Although the government-issued coupon shares similar features with merchant coupon in coupon design, it differs in at least three aspects. First, unlike the merchant coupon that is designed toward promoting a specific product or a particular retailer, the government-issued coupons are nonselective and applicable to almost all types of purchases.² Second, the issuer is government, which may invoke different

psychological process that affects coupon redemption and usage. Third, the government coupon program is launched with the specific policy goal of stimulating consumption during economic downturns, whereas merchant coupons serve different purposes, such as advertising. Previous research finds that merchant coupons can induce brand switch and increase the market share of the promoting brand (see, e.g. Bawa and Shoemaker 1987, Gupta 1988, Blattberg and Neslin 1989). However, the effect of coupons on one's overall spending is unclear. If coupon redeemers simply apply the coupons to routine purchases and effectively turn a government subsidy into saving or increase the spending in a specific product category but decrease purchases in others, then the government coupon program would fail to spur additional spending.

To evaluate the effectiveness of the digital coupon program, we employ a unique and comprehensive data from a large city that is among the first to experiment large-scale digital coupon disbursement. Between March 27 and April 30, 2020, the municipal government rolled out five waves of weekly digital coupons through a mobile payment platform, covering 1.5 million to 2 million individuals in each wave. The total government subsidy reached RMB 500 million. The digital platform allowed one to track the details of coupon acquisition and redemption and hence, to minimize measurement error in estimating consumers' short-term responses to the program.

Using the high-frequency transaction-level data from the digital platform and exploiting a difference-in-differences approach, we estimate the causal effects of coupon acquisition on consumption. Across multiple coupon issuance waves, we find that the treatment group, that is, consumers who acquired coupons, on average spent RMB 121 to RMB 385 more than the control group without coupons during the coupon redemption week. Combined with the actual redemption rate, it suggests that an effective government subsidy of RMB 1 can drive excess spending of RMB 3.4 to RMB 5.8, varying with the design of the coupon. Our analysis also indicates that the expenditure level of the treatment group during the post-coupon weeks is not statistically different from the control group, implying that the increase in spending is not driven by intertemporal substitution. We also find the effectiveness of coupon persists through the multiple waves in our study window.

The large magnitude of excess spending relative to the size of the government subsidy poses a puzzle. Past research shows that a consumer or household typically spends 20 cents to 40 cents for each dollar of tax rebate or government cash payment (see, e.g., Johnson et al. 2006, Parker et al. 2013, Baker et al. 2020). Yet the design of the digital coupon renders a much higher spending tendency. We consider behavioral factors to play a significant role in explaining the

results. In line with mental accounting (Shefrin and Thaler 1981, Thaler 1999), consumers seem to treat the digital coupon as a separate account from their regular weekly budget and exhibit a higher propensity to consume when receiving the small windfall (i.e., the digital coupon packet). The effect is further amplified by loss aversion, which results from the "use-it-or-lose-it" nature of the digital coupons.

This paper provides the first study of the large-scale digital coupon program in China, and our research sheds light on policy design in stimulating consumption. To combat the economic downturn caused by COVID-19, improving the effectiveness of a fiscal stimulus payment program is critical. Our results show that the "small size, short duration" digital coupon program works well in an economy where the households' precautionary savings traditionally are at a high level and the local governments' fiscal budgets are stretched. In addition, the majority of the coupons are redeemed at small businesses, which are most vulnerable to the pandemic shock and in great need of recovery. Because the digital coupons are disbursed through mobile payment platforms, their use can be easily tracked, and the results can be accurately and timely assessed, which allows the policy makers to target the most responsive consumer groups, improve coupon design, and iterate experiment to achieve the intended goal. Our paper is relevant for policy making in the current economic recession, when governments are in an urgent need to bring forward cost-effective policy tools to recover the economy.

Our study is related to a broad literature that investigates consumers' responses to fiscal stimulus payments (see, e.g., Hsieh 2003, Agarwal et al. 2007, Agarwal and Qian 2014, Broda and Parker 2014) as well as the ongoing discussion of the economic impact of the COVID-19 pandemic and the effects of stimulus payments (see, e.g. Baker et al. 2020, Chen et al. 2021). We add to this line of research by using a large-scale, high-frequency, transaction-level data to estimate the effectiveness of an innovative stimulus tool, the digital coupon issued by local governments in China. Voucher programs have been increasingly employed by governments around the world to combat economic downturns. For example, the Singapore government has distributed several rounds of grocery vouchers redeemable at participating merchants and further launched a voucher program to support local tourism business since late 2020. South Korea announced a coupon program to promote consumption in the culture, leisure, and services sectors in October 2020. Our analysis can inform the effects of these programs.

This paper also relates to more recent literature showing that incorporating behavioral considerations into public policy making leads to more desirable policy outcomes (Madrian and Shea 2001, Thaler and

Sunstein 2008, Chetty et al. 2014, Chetty 2015). Sharing a similar spirit, our paper shows that a fiscal stimulus program embedded with features in accordance with behavioral factors such as mental accounting and loss aversion is more effective in stimulating consumption and costs much less.

The rest of the paper proceeds as follows. Section 2 describes the digital coupon program. Section 3 discusses the data and empirical strategy. Section 4 documents the main findings from multiple coupon waves and conducts several robustness checks. Section 5 discusses the potential mechanisms that drive the major findings. The paper concludes in Section 6.

2. The Digital Coupon Program

The COVID-19 pandemic brought much of China’s economy to a standstill with wide lockdowns in the beginning of 2020. According to the National Bureau of Statistics of China, China’s GDP contracted by 6.8% in the first quarter of 2020, the worst growth record in more than three decades, and retail sales of consumer goods and services fell by 19% compared with a year earlier. Chen et al. (2021) reported that the pandemic caused more than 30% of decline in spending on goods and services in China. Even after the spread of virus had been contained and economic activities had been resumed in early March, the recovery of the retail sector was much slower than expected. To kick-start much needed retail spending and support local commerce, local governments started to experiment with the “consumption coupon”.

Our empirical study uses data from Hangzhou, one of the first cities that launched the digital coupon program. Hangzhou is the capital city of the Zhejiang province in the east of China, with 10 million residents

and GDP per capita of RMB 152,465 (approximately US\$22,000). On March 26, the Hangzhou municipal government announced its plan to issue RMB 500 million digital coupons, rolling out in waves from March 27 to April 30, 2020. All residents in Hangzhou (including visitors) are eligible for the coupons. Each person is limited to one claim per wave. The details of coupon design are released one day before each issuance wave.³

The first wave of 2 million coupon packets was released on March 27 at 8 a.m. Each coupon packet includes five separate vouchers of “spend RMB 40, get RMB 10 off” (“RMB 40 -10” thereafter) valid for seven days from March 27 to April 2. The government subsidy is RMB 50 per coupon packet. The coupons can be used with nearly all of the physical merchants in Hangzhou. The second wave of coupons was released on April 3 at 10 a.m., with a total quantity of 1.5 million packets. Each coupon packet includes three different vouchers, one “RMB 100 -20” voucher, one “RMB 200 -35” voucher, and one “RMB 300 -45” voucher. The government subsidy is RMB 100 per coupon packet. The third-wave coupon was rolled out on April 10 at 10 a.m., and the total quantity was again 1.5 million. The coupon design and rules are the same as the first wave, with a government subsidy of RMB 50 per packet.

The digital coupon program was rolled out on Alipay, the dominant mobile payment platform adopted in Hangzhou.⁴ Mobile payment has become ubiquitous for offline purchases through QR code scanning in recent years. When coupons were released, residents at Hangzhou can acquire the coupons by logging onto the Alipay mobile app (Figure 1 shows a screen shot). Simply tap the coupon claim button, and a coupon packet appears in the account and is redeemable immediately. The first release of digital coupons

Figure 1. (Color online) Screenshot of the Coupon Acquisition Page



on March 27 took 38 minutes before all of the 2 million coupon packets were claimed. The time shrank to less than five minutes in all of the subsequent waves.

The vouchers can only be redeemed in physical stores and are not applicable to online transactions as a way to encourage people to go out after a long period of quarantine. The redemption process is easy. When a purchase meets the minimum usage requirement of a voucher, the voucher will be automatically applied to the transaction using Alipay. The convenience of usage partially contributes to the high redemption rate, which is in the range of 70% to 80% across waves.

3. Empirical Strategy and Data

3.1. Empirical Methodology

The main objective of the paper is to evaluate the effect of the government coupon program in the short run. Although the coupons are designed for the consumption context, they are not necessarily effective in generating more spending if consumers apply the coupons toward planned purchase and turn the government subsidy into personal saving. We exploit the fact that only a fraction of Hangzhou residents acquired the government coupons in each wave and employ a difference-in-differences (DID) approach to evaluate the effect of coupon acquisition on individuals' expenditure.

The identification challenge is that coupons are not randomly assigned, and those who obtained the coupon could exhibit different expenditure patterns from those who did not. To alleviate the self-selection concern, we construct the control group as those who attempted but failed to acquire the coupon on the coupon issuance day due to limited offerings. This group of consumers is identifiable, as the digital disbursement of the coupon allows us to track all the visits to the coupon claim page and the activities on the platform. These individuals could potentially be the coupon users and presumably are more similar to the treatment group than those who did not apply for the coupon.

Take the first-wave coupon issued on March 27 as an example. The redemption period is from March 27 to April 2. Our examination window is a 14-day period, which includes a one-week pretreatment period (March 20–26) and a one-week posttreatment or coupon redemption period (March 27–April 2). The treatment effect of coupon acquisition is identified by comparing the consumption level of the treatment group and the control group for the two periods:

$$y_{it} = \alpha_0 + \alpha_1 \text{Treat}_i + \alpha_2 \text{Post}_t + \alpha_3 \text{Treat}_i \times \text{Post}_t + \gamma X_{it} + \varepsilon_{it} \quad (1)$$

The dependent variable y_{it} denotes spending by individual i in period t ($t = 1$, pretreatment period; $t = 2$, posttreatment period). The variable Treat_i is an indicator that is equal to one if individual i is in the treatment

group and zero otherwise, and the coefficient α_1 captures any baseline difference in spending between the treatment and control group. Post_t is a dummy variable indicating whether t refers to the coupon redemption period. The coefficient α_2 captures any time trend in spending across periods. The vector X_{it} controls for other observed individual characteristics that may explain the variation in spending, including gender, age, the average monthly spending over the six months before the program (September 2019 to February 2020), and the total spending in early March (March 1–19). Standard errors are clustered at the individual level.

The coefficient of primary interest is α_3 , which estimates the average excess spending by the treated consumers during the coupon week relative to consumers from the control group. We do not differentiate the number of vouchers redeemed during the week and only consider the average treatment effect of acquiring the digital coupon. The DID framework is able to identify the excess spending (if any) induced by coupon by parceling out the planned or budgeted weekly spending inferred from the control group.

To evaluate the policy efficiency in promoting consumption, we define an efficiency ratio $r = \frac{\alpha_3}{\text{amount of subsidy}}$, where *amount of subsidy* is the average amount of government subsidy per coupon claimer based on the actual redemption. For example, because the coupon receivers in the first wave redeemed 3.5 vouchers ("RMB40 -10" type) per person on average, the average amount of subsidy is RMB 35. The efficiency ratio captures how effective the RMB 1 government subsidy is in driving additional consumption, which reflects the cost-benefit ratio of the policy. This measure is analogous to MPC (marginal propensity to consume), which is defined as the proportion of additional income that individuals spend on consumption. The effectiveness of fiscal stimulus payment depends on MPC.

We further explore the intertemporal effect of coupon acquisition. Specifically, we track a random sample of individuals who received coupon packets in the first wave but not the second wave and extend the observation period to one week after the expiration date of the first wave coupon (April 3–9). The corresponding control group in this analysis is a random sample of consumers who attempted to acquire the coupon on March 27 but did not have the coupon through April 9. We estimate the following model:

$$y_{it} = \alpha_0 + \alpha_1 \text{Treat}_i + \alpha_2 \text{Post}_t + \alpha_3 \text{Post2}_t + \alpha_4 \text{Treat}_i \times \text{Post}_t + \alpha_5 \text{Treat}_i \times \text{Post2}_t + \gamma X_{it} + \varepsilon_{it} \quad (2)$$

The difference between Models (1) and (2) is that we add a postredemption period (April 3–9) to the analysis, which is indicated by the dummy variable Post2_t . Our primary focus is α_5 , which estimates the

Table 1. Summary of Hangzhou Coupon

Wave	Data	Coupon details	Redemption window	Government subsidies (%)	Quantity
I	3/27	5 × “40-10” coupons	7 days	25%	2 million
II	4/3	“100-20”, “200-35”, “300-45”	7 days	16.67%	1.5 million
III	4/10	5 × “40-10” coupons	7 days	25%	1.5 million
IV	4/20	5 × “40-10” coupons	7 days	25%	1.5 million
V	4/30	3 × “40-10” coupons	7 days	25%	3 million

lagged effect of coupon in the subsequent period. If the parameter is negative, it suggests that the treatment group reduced their spending in the subsequent week after the expiration of the coupon relative to the control group. The net effect of coupon acquisition across the two periods is then $(\alpha_4 + \alpha_5)$.

3.2. Data

We use individual-level data from Alipay for the analysis. The basic function of Alipay is an e-wallet, which allows users to transfer money and make payments for both online and offline transactions. Users can link their major bank accounts to the service. In addition, Alipay offers other financial services, including virtual credit card services, *Huabei*, and financial management tools in the app. We have access to the de-identified account level transaction details for a total number of 1 million individuals sampled in this study.⁵

For each individual in the sample, we observe the complete transaction information of the account and the account holder’s personal attributes, such as gender and age. The transaction-level data include the transaction time, transaction amount, the usage of coupon (if any), merchant category, and whether the transaction is online or offline. The advantage of this

data set is that we have a complete record of coupon acquisition and redemption with little measurement error, whereas previous studies of government shopping coupon programs rely on survey and secondary data to infer the effectiveness of coupon issuance (Hsieh et al. 2010, Kan et al. 2017).

In each wave of coupon disbursement, 1.5 million to 3 million individuals successfully obtained the coupon packets (see Table 1), and 1 million to 2.7 million users attempted but failed to acquire the coupon on the coupon issuance day. To evaluate the effect size of a specific coupon wave, we take a random sample of 100,000 users from the coupon receivers in that wave to construct the treatment group and 100,000 users who attempted but failed to obtain the coupon on the same day as the control group. For the purpose of our study, we aggregate the individual daily transactions to a weekly level, where a “week” is defined as a seven-day period before or after a specific coupon issuance day, that is, the pretreatment and posttreatment (redemption) period. Therefore, in each coupon wave analysis, our sample includes 200,000 individuals with pre- and posttreatment spending observations. Our study focuses on the first two waves. The sample summary statistics of Wave I and Wave II are reported in Table 2. To minimize the influence of

Table 2. Summary Statistics

	Full sample		With redemption		Without redemption	
	Mean	SD	Mean	SD	Mean	SD
Panel A: Wave I						
Gender	0.508	0.500	0.448	0.497	0.495	0.500
Age	34.631	10.483	34.736	10.104	32.353	10.068
Spending in March	3,163.430	6,697.497	3,567.234	6,910.231	2,543.555	5,458.002
Average monthly spending	6,508.314	9,231.686	7,366.879	9,207.443	5,430.750	7,296.476
Obs.	200,000		88,332		11,668	
Panel B: Wave II						
Gender	0.476	0.499	0.415	0.493	0.488	0.500
Age	35.711	10.854	36.385	10.679	32.914	10.384
Spending in March	3,142.003	6,643.242	3,455.496	7,028.309	2,773.745	5,924.238
Average monthly spending	6,420.796	8,846.596	7,016.254	9,260.108	5,625.187	7,762.301
Obs.	200,000		81,352		18,648	

Notes. Gender is a dummy that equals one for male and zero for female, *Spending in March* means the total spending from March 1 to March 19 in 2020, and *Average monthly spending* means the average monthly spending over the six months before the coupon program (from September 2019 to February 2020). The unit of spending is in RMB yuan. “With redemption” refers to the group of individuals that redeemed at least one voucher in a coupon wave and “Without redemption” refers to the group of individuals that successfully claimed but did not redeem any of the vouchers.

the outliers, we winsorize the variables of spending amount at the 99.5 percentile.

The first-wave coupon packet includes five “spend RMB 40, get RMB 10 off” vouchers. About 88.3% of the coupon claimers used at least one voucher during the redemption period. The individuals in the experiment group (including the ones with no coupon redemption) on average redeemed 3.5 coupons during the seven days when the coupons were valid, which means the government subsidized approximately RMB 35 per coupon claimer. The second-wave coupon packets include three different vouchers, that is, “RMB 100 -20”, “RMB 200 -35”, and “RMB 300 -45”. The redemption rates for each are 68.5%, 61.4%, and 63.5%, respectively. On average, the government subsidized RMB 63.8 per coupon claimer. Table 2 shows the comparison between the individuals who redeemed at least one voucher and those who successfully claimed but did not redeem any voucher. We find that the group with coupon redemption has a larger fraction of female, higher spending in the past and is slightly older than the group that claimed but did not use coupon.

In Table 3, we report separately the statistics of the treatment and control groups in Wave I and Wave II.

We notice that there exists differences between the two groups. For example, the control group exhibits lower spending on average than the treatment group. To address the issue, we use the propensity score matching method (PSM hereafter) to construct a more comparable sample of those who successfully obtained the coupon and those who failed. The variables used for matching include age, gender, the average monthly spending in the last six months, the average number of merchant categories purchased each month in the last six months, the total spending in early March (March 1–19), and whether one has a *Huabei* account.⁶ For each individual, we compute the propensity score based on a logistic regression with the above variables and perform the nearest-neighbor matching. We drop the observations that cannot be matched; that is, the gap of propensity scores exceeds the pre-specified threshold. After matching, the differences between the treatment and control group are considerably reduced and almost indistinguishable.

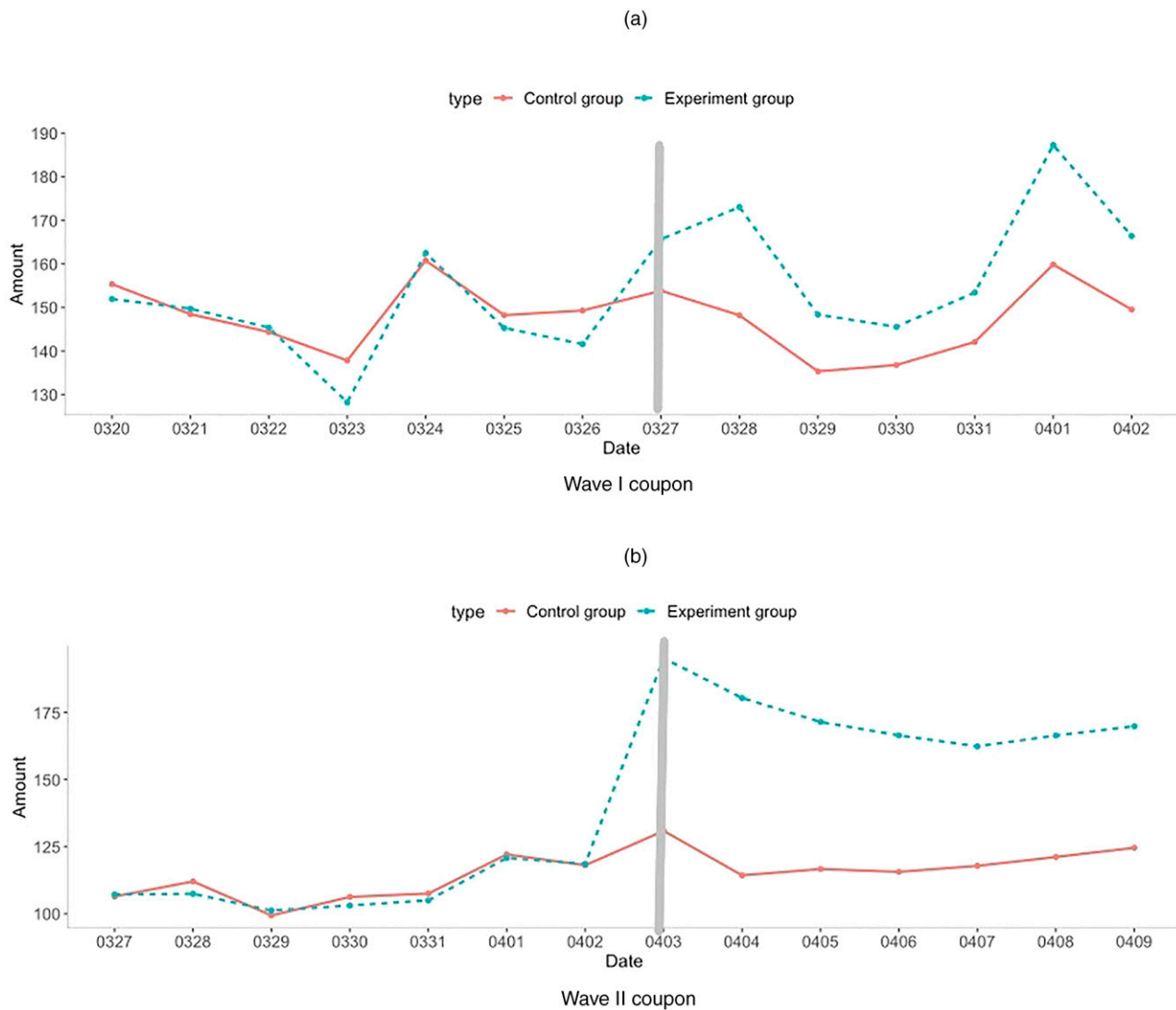
As model-free evidence for the effectiveness of the coupon program, we show in Figure 2 the time-series plots of the average daily spending of the treatment group and the control group of the matched sample

Table 3. Comparison Between the Treatment Group and Control Group

	Treatment group		Control group		p-value
	Mean	SD	Mean	SD	
Panel A: Wave I					
Before matching					
Gender	0.454	0.498	0.562	0.496	0.000
Age	34.458	10.128	34.805	10.823	0.000
Spending in March	3,447.791	6,764.864	2,879.068	6,617.255	0.000
Average monthly spending	7,140.971	9,026.803	5,875.656	9,389.612	0.000
Obs.	100,000		100,000		
After matching					
Gender	0.512	0.500	0.516	0.500	0.138
Age	34.566	10.818	34.561	10.577	0.186
Spending in March	3,144.434	6,401.660	3,195.562	6,880.953	0.453
Average monthly spending	6,459.872	8,529.116	6,548.897	9,605.266	0.033
Obs.	67,742		67,742		
Panel B: Wave II					
Before Matching					
Gender	0.429	0.495	0.505	0.500	0.000
Age	35.737	10.710	35.685	10.996	0.123
Spending in March	3,328.363	6,841.103	2,955.644	6,433.926	0.000
Average monthly spending	6,756.848	9,016.002	6,084.745	8,660.874	0.000
Obs.	100,000		100,000		
After Matching					
Gender	0.468	0.499	0.467	0.499	0.697
Age	35.752	11.022	35.739	10.906	0.760
Spending in March	3,099.724	6,536.685	3,095.629	6,532.143	0.863
Average monthly spending	6,391.641	8,878.233	6,346.141	8,583.230	0.152
Obs.	75,501		75,501		

Notes. *Gender* is a dummy that equals one for male and zero for female, *Spending in March* means the total spending from March 1 to March 19 in 2020, and *Average monthly spending* means the average monthly spending over the six months before the coupon program (from September 2019 to February 2020). The unit of spending is in RMB yuan.

Figure 2. (Color online) Time Trend of Spending



Note. The solid line represents spending by the control group, and the dotted line represents spending by the treatment group.

for the period before and after Wave I and Wave II coupon issuances. The figure shows that the two groups followed a similar trend in total spending before the coupon rollout, and there is an immediate surge of spending by the treatment group following the receipt of coupons on the issuance day. It provides suggestive evidence that the government-issued coupons effectively induced excess spending.

4. Main Results

4.1. The Short-Term Effect of Coupon Acquisition

We first report the average effect of coupon acquisition on individual spending during the redemption window, which estimates the immediate effect of government coupon issuance. Table 4 presents the DID analysis of the first-wave coupon using Equation (1).

We performed the analysis on both the full-sample (Columns (1)–(3)) and the PSM-matched sample (Columns (4)–(6)).

The estimated coefficient of the interaction term $Treat \times Post$ measures the treatment effect during the seven-day coupon redemption window. Column (1) shows that individuals in the treatment group on average increased their total spending by RMB 124.6 in the coupon week relative to the controls, and the estimate using the matched sample in Column (4) is similar, RMB 144.1. The effect is both statistically and economically significant. It means that a government subsidy of RMB 35 per coupon claimer (considering the actual redemption rate) induced RMB 124.6 worth of unplanned spending (using the full sample estimates), which corresponds to an efficiency ratio of 3.5.

Table 4. The Average Effect on Spending (Wave I)

	Full sample			PSM sample		
	(1) Payamt	(2) Offline	(3) Online	(4) Payamt	(5) Offline	(6) Online
Treat × Post	12,464.380*** (888.192)	14,553.120*** (639.475)	−2,342.192*** (300.298)	14,410.720*** (1,092.934)	15,396.330*** (792.212)	−1,063.490*** (366.306)
Treat	931.611 (784.341)	−1,131.194* (586.683)	3,811.373*** (307.666)	−3,732.973*** (924.151)	−1,980.108*** (690.976)	−1,918.443*** (362.071)
Post	−1,138.626* (611.138)	755.695* (441.759)	−1,723.784*** (202.863)	−2,274.322*** (777.930)	150.261 (563.241)	−2,210.782*** (264.543)
Age	−110.570 (172.050)	1,315.736*** (132.558)	−634.253*** (69.876)	−18.131 (202.485)	1,347.798*** (161.111)	−814.739*** (81.270)
Age ²	−2.608 (2.187)	−10.687*** (1.717)	−1.387 (0.871)	−3.631 (2.573)	−12.415*** (2.084)	−0.644 (1.012)
Gender	4,984.395*** (612.942)	14,409.530*** (464.594)	−10,644.580*** (251.976)	6,765.630*** (738.036)	15,598.660*** (562.391)	−9,586.137*** (304.124)
Payamt_3up	0.073*** (0.002)	0.040*** (0.001)	0.017*** (0.001)	0.071*** (0.002)	0.040*** (0.002)	0.015*** (0.001)
Payamt_6m	0.089*** (0.001)	0.051*** (0.001)	0.023*** (0.0004)	0.089*** (0.001)	0.053*** (0.001)	0.021*** (0.001)
Constant	25,590.810*** (2,977.548)	−29,585.610*** (2,237.926)	44,923.290*** (1,244.877)	26,551.090*** (3,510.465)	−31,532.160*** (2,695.381)	51,759.330*** (1,469.060)
Obs.	400,000	400,000	400,000	270,968	270,968	270,968
Adj. R ²	0.324	0.218	0.181	0.311	0.222	0.162

Notes. Standard errors (in parentheses) are clustered at the individual level. *Payamt_3up* means the total spending from March 1 to March 19 in 2020, and *Payamt_6m* is the average monthly spending over the six months before the coupon program (from September 2019 to February 2020). The unit of spending is in cents (0.01 RMB yuan).

***1%, **5%, and *10%, statistical significance.

Columns (2) and (3) decompose the total spending into offline spending and online spending. The results indicate that the offline payment increased by RMB 145.5, whereas the online spending was reduced by RMB 23.4 during the coupon week. This is not surprising, because the coupon program is aimed at stimulating local business and the coupons are applicable to offline purchases only. However, the substitution from the online channel is relatively small. The analysis using the PSM sample yields similar results as presented in Columns (5) and (6).

Table 5 reports the main effect of the second wave coupon rolling out on April 3. As shown in Column (1), the average treatment effect is RMB 372.6 for the full sample. That is, the individuals with the coupon on average spent RMB 372.6 more than their counterparts in the control group during the coupon redemption week of the second wave. Considering that government effectively subsidized RMB 63.8 per coupon packet in the second wave, the estimated efficiency ratio is 5.8. The ratio higher than the first wave is due possibly to the fact that the required minimum spending to redeem a voucher is higher in the second wave, which leads to a larger purchase amount. The results from the matched sample analysis are highly consistent.

Table 6 reports the category-level analysis for the first- and second-wave coupon, respectively, using the matched samples. In both waves, we find that there is

a substantial increase in expenditure on restaurants and food and drinks for the treatment group relative to the control group during the coupon redemption week. The incremental consumption from these two categories is RMB 69.5 (RMB 51.6 + RMB 17.9) in Wave I and RMB 179.8 (RMB 124.2 + RMB 55.6) in Wave II, which contributes to roughly half of the coupon effect. Our results suggest that many consumers respond to the small-value coupon with more spending in daily consumption such as eating and drinking, which typically involves low transaction amounts. It also suggests that the digital coupon program largely benefits the catering service sector, which was most affected by the outbreak of COVID-19.⁷

4.2. The Dynamic Effect of Coupon Issuance

The results from both coupon waves indicate that coupon receivers significantly increased their spending during the redemption week relative to their counterparts in the control group. An important question is whether such coupon-induced extra expenditure would crowd out consumption in the subsequent period, or in other words, whether the treatment effect of coupon is due to intertemporal substitution.

To examine the hypothesis, we draw a random sample of 100,000 individuals who received coupon packets in the first wave but not the second wave and track their purchases till one week after the coupon expiration date,

Table 5. The Average Effect on Spending (Wave II)

	Full sample			PSM sample		
	(1) Payamt	(2) Offline	(3) Online	(4) Payamt	(5) Offline	(6) Online
Treat × Post	37,263.840*** (876.473)	38,798.470*** (655.791)	−1,198.595*** (269.603)	37,970.600*** (1,009.633)	39,081.760*** (758.004)	−939.177*** (309.814)
Treat	6,105.498*** (738.997)	5,746.848*** (563.469)	624.017** (265.235)	4,457.775*** (841.554)	4,842.247*** (648.489)	−750.347** (299.310)
Post	−1,133.237* (603.148)	−546.219 (448.597)	−280.054 (187.000)	−845.265 (707.974)	−536.332 (528.659)	−206.208 (219.332)
Age	599.328*** (173.960)	1,735.869*** (135.775)	−494.230*** (62.627)	542.245*** (194.029)	1,792.664*** (150.332)	−692.667*** (70.463)
Age ²	−9.114*** (2.216)	−14.194*** (1.747)	−2.777*** (0.776)	−8.779*** (2.451)	−15.015*** (1.917)	−0.525 (0.870)
Gender	8,766.232*** (619.096)	14,966.440*** (493.636)	−8,028.829*** (222.473)	9,362.864*** (697.030)	14,874.140*** (553.154)	−6,812.374*** (252.782)
Payamt_3up	0.067*** (0.002)	0.041*** (0.001)	0.013*** (0.001)	0.067*** (0.002)	0.040*** (0.002)	0.013*** (0.001)
Payamt_6m	0.092*** (0.001)	0.059*** (0.001)	0.019*** (0.0004)	0.092*** (0.001)	0.061*** (0.001)	0.018*** (0.0004)
Constant	11,279.330*** (3,048.013)	−37,549.190*** (2,325.164)	41,330.150*** (1,138.730)	13,531.760*** (3,482.019)	−38,328.020*** (2,651.070)	45,639.310*** (1,296.434)
Obs.	400,000	400,000	400,000	302,004	302,004	302,004
Adj. R ²	0.318	0.243	0.153	0.311	0.243	0.144

Notes. Standard errors (in parentheses) are clustered at the individual level. *Payamt_3up* means the total spending from March 1 to March 19 in 2020, and *Payamt_6m* is the average monthly spending over the six months before the coupon program (from September 2019 to February 2020). The unit of spending is in cents (0.01 RMB yuan).

***1%, **5%, and *10%, statistical significance.

Table 6. Cross-Category Analysis

Panel A: 3/27 Wave I					
	(1) Restaurant	(2) Food	(3) Clothes	(4) Nondurable	(5) Other service
Treat × Post	5,157.611*** (251.703)	1,787.772*** (47.005)	319.355*** (72.469)	84.398*** (9.424)	389.965* (235.305)
Treat	−1,366.927*** (224.460)	155.938*** (47.805)	−113.773* (68.174)	−23.869*** (7.184)	−1,931.552*** (206.774)
Post	−193.555 (180.346)	−71.143** (31.281)	−1,096.514*** (51.269)	−25.138*** (6.508)	−252.671 (171.205)
Constant	30,527.060*** (947.824)	−3,398.959*** (225.309)	6,370.993*** (254.669)	26.162 (29.431)	12,482.280*** (829.253)
Controls	Yes	Yes	Yes	Yes	Yes
Obs.	270,968	270,968	270,968	270,968	270,968
Adj. R ²	0.159	0.068	0.070	0.007	0.118
Panel B: 4/3 Wave II					
	(1) Restaurant	(2) Food	(3) Clothes	(4) Nondurable	(5) Other Service
Treat × Post	12,422.110*** (246.595)	5,556.366*** (69.172)	2,153.419*** (75.363)	66.311*** (10.621)	2,170.140*** (211.930)
Treat	1,230.344*** (204.738)	873.939*** (56.051)	253.976*** (57.689)	17.453** (8.235)	−692.891*** (179.399)
Post	−713.625*** (166.916)	−743.842*** (36.627)	446.336*** (47.823)	−9.162 (7.203)	30.905 (148.274)
Constant	26,409.580*** (948.367)	−6,755.225*** (288.247)	4,586.626*** (269.301)	30.162 (34.715)	11,239.020*** (732.269)
Controls	Yes	Yes	Yes	Yes	Yes
Obs.	302,004	302,004	302,004	302,004	302,004
Adj. R ²	0.156	0.086	0.058	0.007	0.107

Notes. Standard errors (in parentheses) are clustered at the individual level. The unit of spending is in cents (0.01 RMB).

***1%, **5%, and *10%, statistical significance.

Table 7. The Lagged Effect Estimation

	(1) Wave I	(2) Wave V
Treat × Post	10,388.050*** (1,072.081)	12,134.050*** (967.419)
Treat × Post2	−1,108.379 (1,111.126)	−954.153 (972.522)
Treat × Post3		405.610 (991.760)
Treat × Post4		533.170 (978.560)
Treat × Post5		1,856.725* (1,022.202)
Treat	2,745.210*** (800.195)	−314.001 (844.581)
Post	−918.721 (736.508)	5,588.991*** (679.024)
Post2	350.264 (766.683)	−793.486 (693.048)
Post3		3,082.770*** (709.009)
post4		−6,236.438*** (700.490)
post5		7,100.142*** (733.301)
Constant	23,570.630*** (2,842.396)	14,408.630*** (2,593.344)
Controls	Yes	Yes
Obs.	600,000	1,200,000
Adj. R ²	0.276	0.302

Notes. Standard errors (in parentheses) are clustered at the individual level. The unit of spending is in cents (0.01 RMB yuan).

***1%, **5%, and *10%, statistical significance.

that is, April 3 to April 9. The control group includes 100,000 individuals who attempted but failed to obtain coupons during the whole observation window from March 27 to April 9. We estimate the lagged effect of coupon issuance using Equation (2) and report the results in Column (1) of Table 7. The results indicate that the experiment group on average spent RMB 103.9 more than the control group during the coupon week and spent RMB 11.1 less during the post-coupon week. However, the reduction in the post-coupon period is statistically insignificant. It suggests that the excess spending incurred during the coupon redemption period does not significantly crowd out subsequent spending of the week after.

A potential concern is that the substitution effect may occur in the longer term. To this end, we utilize data from the last coupon wave rolling out on April 30 (wave V) to further check the lagged effect. This wave offers a coupon packet with three “Spend RMB 40, get RMB 10 off” vouchers with a seven-day redemption period. No other coupons were issued in the subsequent weeks. We take a random sample of 100,000 coupon receivers and 100,000 individuals who attempted but failed to get coupon in that wave and track their consumption behavior for four weeks after the coupon redemption period. We estimate an

extended version of Equation (2) with five posttreatment periods and report the results in Column (2) of Table 7. We find that although the treatment group spent more than the control group during the coupon redemption week, as indicated by the positive and significant interaction term of *Treat × Post*, the spending difference is negligible and statistically insignificant for the weeks beyond the redemption period. It suggests again that there is little evidence of consumption crowd-out.

Another question is whether the effectiveness of coupons in inducing excess spending wears out easily given the high frequency of coupon rollout. Therefore, we examine the third-wave coupon rolling out on April 10, which has the exact design as the first wave. The estimated treatment effect is about RMB 139, which is quite close to the estimate from the first wave.⁸ Given that the redemption rate in this wave is 82.4%, it means that RMB 1 of government subsidy leads to RMB 3.4 excess spending, which is also highly comparable with the estimated ratio 3.5 in the first wave. In addition, the analysis of the last-wave coupon above shows that the treatment effect is RMB 121.3, as indicated in Column (2) of Table 7. Given that the redemption rate in this wave is close to 80%, the estimated policy efficiency ratio is 5.1.

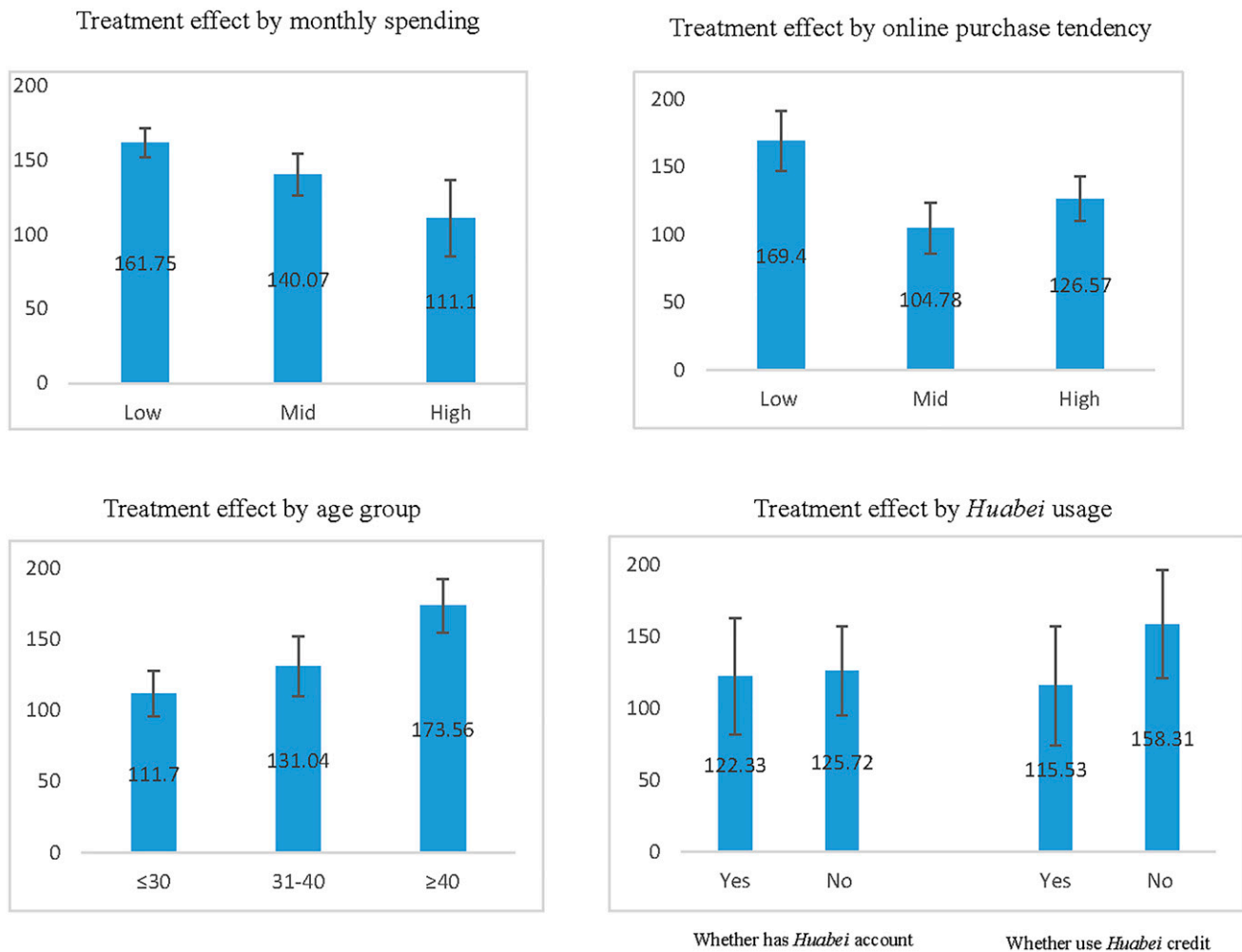
Taking the evidence together, we find that the week-long coupon program can effectively increase the spending during the redemption period without crowding out the subsequent purchases. In addition, the coupon effect in boosting spending persists across multiple rounds of coupon issuance. One possible reason is that the excess spending induced by the coupon concentrates on high-purchase frequency items such as eating and drinking. Unlike purchases of durable goods, expenditure in such categories in the current period has little influence on the future spending.

4.3. Heterogeneity Across Consumer Groups

We further study how the effectiveness of coupons in stimulating spending varies across sections by exploiting the fact that we observe account holders’ demographic characteristics and past transactions. Figure 3 displays the estimated coupon treatment effect for different consumer groups based on the analysis of the first-wave coupon.⁹

First, we split our sample into terciles according to an individual’s average monthly expenditure during the six months before the program and run a separate DID analysis for each tercile. The estimates indicate that the low-expenditure group is the most responsive to the government coupon. Similarly, we divide the sample based on consumers’ tendency to purchase online versus offline. For this, we construct an online index that is computed as the ratio of the total amount of online transactions in the last six months relative to the

Figure 3. (Color online) Heterogenous Treatment Effect by Consumer Groups



total transaction amount during the same period of time. We find that the treatment effect is the most pronounced among the group with low online index, which suggests that consumers who are used to offline channels for purchases are more responsive to the coupon program. We also analyze how different age groups respond differently to the program and find that consumers aged above 40 are the most responsive.

Another dimension we consider is whether one has access to *Huabei*, the virtual credit card service provided by Alipay, which may reflect one's need for credit. However, we do not find the treatment effect of coupons to differ for *Huabei* users and nonusers. We further compare the effects within *Huabei* users by whether one has ever used the granted credit in transaction in the last six months, which could be an indicator of liquidity constraint. We find that the consumers who have not used the granted credit before are more responsive to the coupons. The evidence seems to suggest that liquidity constraint is an unlikely explanation for the high responsiveness to the coupon program.

4.4. Robustness Checks

We perform several additional analyses to check the robustness of the results. The first concern is that the treatment group and the control group might be different along unobserved dimensions. For example, because the coupons were distributed in the way of "first come first serve", those who arrived early and therefore successfully acquired the coupon might have a higher spending propensity than the control group, although the coupons are typically all claimed within minutes in each wave. To test whether the timing of coupon application is correlated with the responsiveness, we split the treatment group evenly into two halves, *Early* and *Late*, where *Early* means the first half based on the time of arrival in coupon application. If the argument is true, then we should see a different treatment effect across the two groups. Yet the estimation results in Panel A of Table 8 using data from the Wave I coupon do not support the argument. The effect size difference of the two groups is statistically insignificant, which suggests that the order of

Table 8. Robustness Checks

Panel A: Split sample			
TreatE × Post	12,354.210***		
	(895.164)		
TreatL × Post	13,406.640***		
	(899.313)		
TreatE	241.734		
	(786.075)		
TreatL	712.491		
	(794.028)		
Post	−1,552.671**		
	(614.775)		
Constant	28,867.140***		
	(2,618.740)		
Controls	Yes		
Obs.	600,000		
Adj. R ²	0.318		
Panel B: Restricted samples			
	(1) 10-minute window	(2) 3-minute window	(3) Spending threshold
Treat × Post	13,468.300***	43,820.360***	12,567.310***
	(1,889.157)	(1,224.212)	(959.884)
Treat	591.576	2,831.361***	−752.351
	(1,639.370)	(1,066.778)	(830.874)
Post	−202.214	8,267.601***	−1,363.686*
	(1,321.028)	(838.255)	(696.542)
Constant	24,057.170***	13,849.280***	28,828.420***
	(6,284.944)	(4,627.067)	(3,309.497)
Controls	Yes	Yes	Yes
Obs.	100,000	200,000	369,528
Adj. R ²	0.338	0.323	0.314

Notes. Standard errors (in parentheses) are clustered at the individual level. The unit of spending is in cents (0.01 RMB).

***1%, **5%, and *10%, statistical significance.

arrival in coupon application is not correlated with coupon responsiveness.

As a further test, we reconstruct the treatment and control group by restricting the sample to the individuals who arrived at the coupon claim page during a short time window before and after the moment that all the coupons were claimed. Note that this moment is unknown ex ante. Presumably, the individuals who arrived just before the all-claimed moment and successfully obtained the coupon are not different from the group who barely missed it. For the robustness check, we use the time window of 10 minutes in Wave I. To form the control group, we randomly draw 25,000 individuals among the roughly 28,000 users that arrived within 10 minutes after the all-claimed moment and failed to get the coupon. The treatment group is a random sample of 25,000 individuals who successfully acquired the coupon within 10 minutes before the all-claimed moment. The estimated treatment effect is RMB 134.7 as reported in Column (1) of Panel B, which is highly consistent with the estimate RMB 124.6 obtained from the full sample. We repeat the same analysis using time window of 3 minutes on Wave II and display the results in Column (2). The estimated

treatment effect on this sample is RMB 438.2, which is slightly higher than the full sample estimate.

Another concern is that consumers with coupons may shift their transactions from other payment methods to Alipay in order to redeem the coupon. To partially address the concern, we run the analysis by excluding individuals with low transaction amounts (less than RMB 100 per month) before the coupon program, because this group is more likely to rely on other payment methods, and switching is more likely to occur. Column (3) of Panel B reports the results using data from the first-wave coupon. We find that the estimated treatment effect using the restricted sample is not statistically different from that of the full sample analysis. It implies that substitution from other payment methods is not the main reason for the increased consumption. In addition, our survey conducted in the middle of April among the local residents shows that one's choice of payment method is relatively stable and is not affected by coupon usage.

Overall, these analyses show the robustness of the main findings. In addition, to check whether the main results extend beyond the context of Hangzhou, arguably one of the most developed areas in China, we

obtained data of a similar digital coupon program from the Guangxi province, located in less developed southern China. The empirical analysis using the same methods reveals that a government subsidy of RMB 1 spurred RMB 5.2 excess spending in that region, which suggests that the digital program is also effective in driving additional consumption in less developed regions.¹⁰

5. Potential Explanations

Our analysis of the Hangzhou digital coupon program shows that the government-issued coupon is highly effective in stimulating consumption. An effective government subsidy of RMB 1 can drive excess spending of RMB 3.4 to RMB 5.8. This large magnitude of effect size poses a challenge to standard explanations. First, our estimation is not consistent with the life cycle/permanent income (LCPI) model (Modigliani and Brumberg 1954, Friedman 1957). According to the LCPI theory, these coupons are too small in size and should have inconsequential effects on spending. Also, a liquidity constraints hypothesis can hardly explain the results for the same reason.

One explanation for the spending increase is the price effect. The coupon essentially offers a price discount of a transaction. Extensive research in economics and marketing has studied the coupons issued by manufacturers or merchants (see, e.g., Narasimhan 1984, Varian 1989). The merchant coupon can increase short-term sales mainly from two sources. One is due to the switching effect that consumers switch from the competitors to the merchant as a result of the coupon (see, e.g. Shaffer and Zhang 1995), and the other is sales from new customers or a market expansion effect (see, e.g., Reimers and Xie 2019). However, in our setting, the coupons are issued by the government and applicable to all of the merchants. The switching effect cannot explain an increase in spending at the aggregate level. The market expansion effect is more relevant. That is, a consumer is unwilling to make a purchase under the regular price but is willing to buy with the coupon discount. However, consumers in our setting have the option to apply the coupon in routine purchases and save money rather than incurring additional expenditure. Therefore, we argue that the price effect alone is unlikely to be the driving force of the coupon-induced excess purchases.

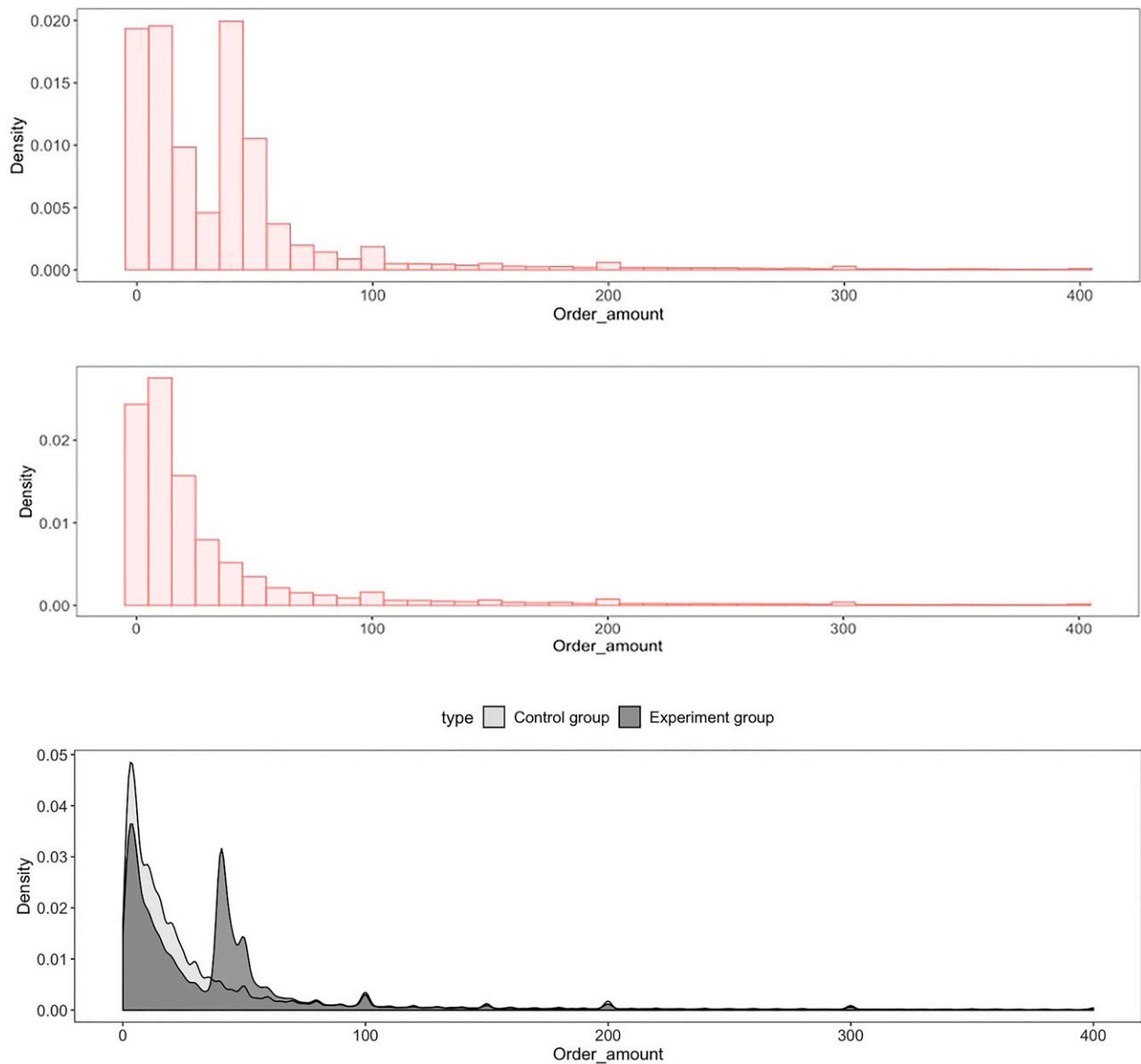
We turn to behavioral factors to investigate the potential mechanisms operative. One potential factor is mental accounting (Shefrin and Thaler 1981, Thaler 1985, 1999). The mental accounting explanation suggests that consumers treat the digital coupon as part of a separate account. That is, income is nonfungible in the sense that it is divided into categories (e.g., regular weekly budget versus coupon-related income account), and consumers spend within the budget.

The coupon account involves a small windfall, and a consumer may use this unexpected income to purchase items he or she would not purchase otherwise (Milkman and Beshears 2009). That is, the propensity to consume is higher under the coupon account than the regular account.

As a test of the null hypothesis of fungibility motivated by mental accounting, we examine the distribution of the consumers' purchase size. If the money in the coupon-related expenditure account will be spent just as well as in another, the distribution of the treatment group's purchase size should not be significantly different from that of the control group, as the coupon can be applied to any transaction above a threshold. The top two panels of Figure 4 present the histogram of the purchase size (from 0 to RMB 400) for the treatment and control groups in Wave I, respectively. It is highly noticeable that the purchase size of the treated consumers clusters at RMB 40, whereas we do not find the same pattern for the control group. The bottom panel of Figure 4 depicts the kernel distribution of the two groups. The Kolmogorov-Smirnov test rejects the null hypothesis that the two distributions are equal. Figure 5 shows the same analysis for the second wave. Again, the evidence suggests that coupon-related expenditure is not treated as the usual spending account.

Nonetheless, mental accounting alone cannot fully rationalize the large magnitude of coupon effect ranging from 3.5 to 5.8. We speculate that the large effect size in our setting, an order of magnitude larger than those in previous studies, has a lot to do with the coupon program's several salient features. These distinctive features are built on behavioral insights that influence behavior, among which loss aversion generates a first-order effect.¹¹ Loss aversion emerges from a behavioral model with reference-dependent preference (Kahneman and Tversky 1979). There are many examples of the loss aversion effect in both the laboratory and field experiments (see, e.g., Odean 1998, List 2003). In this setting, the coupon subsidy is not cash equivalent and cannot be captured unless the spending reaches a certain amount. The tendency to avoid "loss," that is, not claiming the government subsidy, incentivizes coupon holders to spend. The implementation of the coupon rollout, that consumers have to compete for a limited supply of the coupons (albeit in large quantity), further amplifies loss aversion. The scarcity effect (Cialdini 1993) may increase the perceived value of the coupons among those who have successfully acquired.

Notably, loss aversion alone cannot explain the high tendency to spend with coupon either. In the absence of mental accounting, the coupon subsidy could be fungible with other source of income, and coupon-induced spending may well substitute for consumers'

Figure 4. (Color online) Distribution of Purchase Size - Wave I

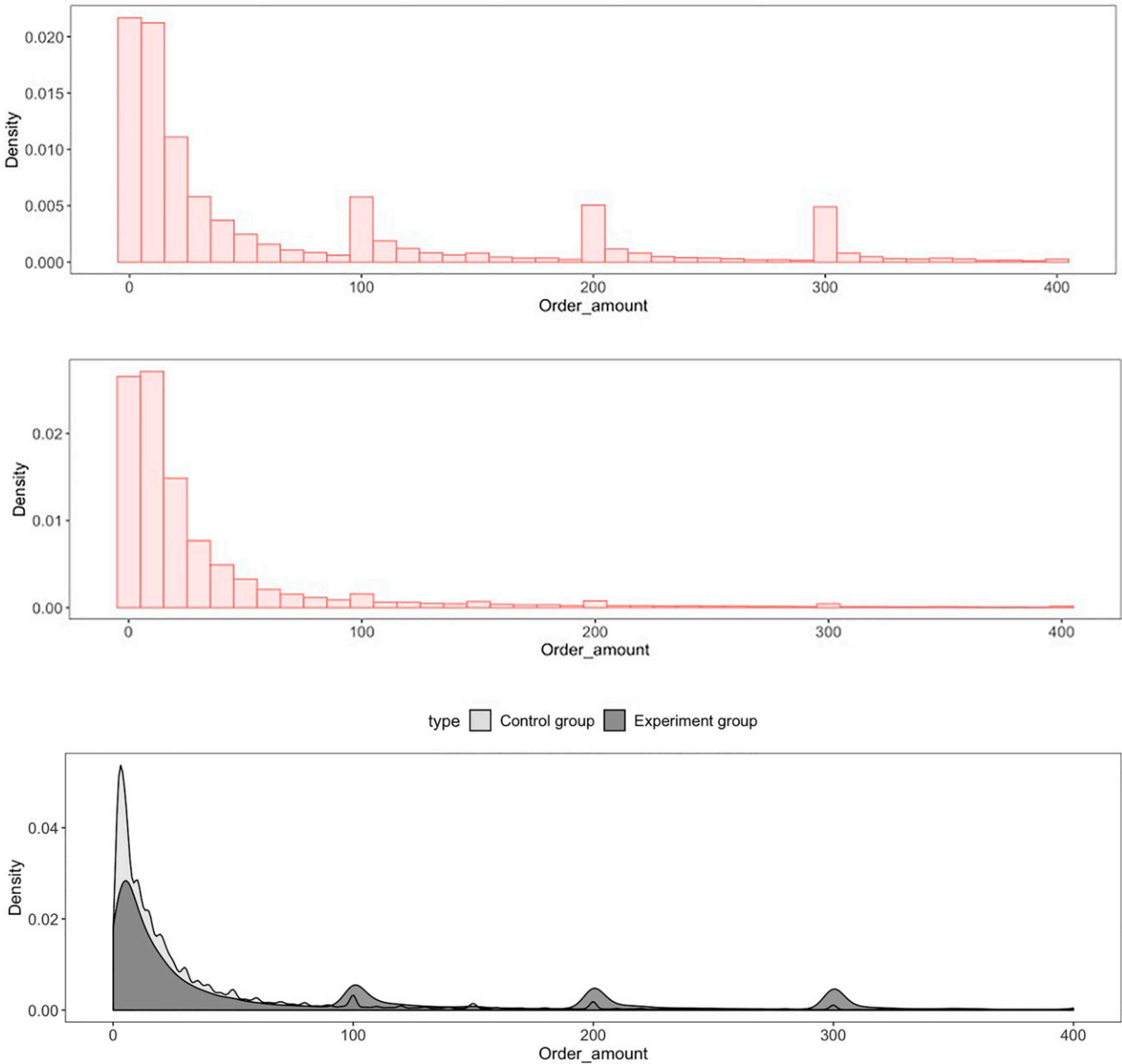
regular weekly expenditure, in which case we would fail to see coupon holders spend significantly more than the nonholders. Mental accounting and loss framing are two primary mechanisms driving higher effect size in our study.

There are other behavioral factors that may contribute to the effect. First, by naming the vouchers as “consumption coupon” (a direct translation from Chinese), it highlights spending as the stated goal of the government subsidy and helps to enhance the effect. This can be considered as a sort of framing or labeling effect (see, e.g., Kooreman 2000, Benhassine et al. 2015). In addition, the digital coupon is issued by the government and promoted as a way to support local business on news media, which may

trigger other psychological responses such as reciprocity (Rabin 1993) that further enhance the effectiveness of the coupon program. It is notable that the redemption rate of the coupon across waves is substantially higher than that of the merchant coupon promotions reported in the literature (see, e.g., Sahni et al. 2017).

We conjecture that the effectiveness of the coupon program is a function of consumers’ awareness of these behavioral biases and whether they learn to counteract with the biases (List 2003). As reported in Section 4.3, we find that consumers with lower income (expenditure) and fewer online shopping experiences and in older age groups tend to be more responsive to the coupon program. One interpretation of this

Figure 5. (Color online) Distribution of Purchase Size - Wave II



finding is that these consumers are less experienced with digital coupons and more subject to the influence of behavioral factors.

6. Conclusion

This paper studies a large-scale digital coupon program by local governments in China to stimulate consumption in the pandemic. Our analysis of multiple waves of the digital coupon program utilizing a rich panel of transaction-level data of more than 1 million individuals yields several notable findings. (1) Consumers are highly responsive to the small-value, short-duration coupon issued by the government. An RMB 1 government subsidy can generate RMB 3.4 to

RMB 5.8 excess spending.¹² (2) The increase in consumption during a coupon redemption week does not negatively affect future spending, and the effectiveness of coupon does not wear out over waves of issuance. The incremental consumption induced by the coupons largely falls in everyday purchases such as food and catering and hence, benefits small businesses. (3) There is substantial heterogeneity in coupon responsiveness across different consumer groups, which could be related to behavioral factors.

Our research has important policy implications. It suggests that a well-designed coupon program can be a useful policy tool to stimulate consumption, which complements the usual stimulus programs such as

cash payment and tax cut in response to economic downturn. The experimentation of such programs is particularly relevant when governments are in an urgent need to bring forward cost-effective policies to recover the economy. We find that the small-valued coupons directly benefit the local small businesses, which have attracted a high proportion of coupon redemption, and there could be indirect effects from the program that benefit the entire local economy as consumers resume their shopping trips. In addition, by breaking a usually one-time and large-scale nationwide program into many localized programs, the policy makers and economists can better evaluate the first-order impacts of those critical policy details and pursue good policy implementation.

Finally, our research sheds light on the broader discussion of policy design integrated with behavioral insights (Chetty 2015, Duflo 2017). The high effectiveness of the digital coupons as documented is closely related to the design and implementation of the program, which invokes several behavioral mechanisms of human behavior. Future research is needed to fully understand the underlying mechanisms and potentially decompose the overall effect. The behavioral insights, coupled with digital technology that enables real-time tracking and evaluation, offer prominent opportunities to explore more effective public policy tools.

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Endnotes

¹ Exceptions include Japan's shopping coupon program in 1999 and Taiwan's shopping voucher program in 2009. These coupons are cash-equivalent and different from the coupons in our context. Hsieh et al. (2010) found that Japan's shopping coupons in 1999 induced a modest marginal propensity to consume (MPC) on semi-durables in the magnitude of 0.1–0.2. Kan et al. (2017) use survey data to investigate consumers' responses to the Taiwan shopping voucher program and find an MPC of 0.24.

² In this regard, the government-issued coupon in our context also differs from other government voucher programs that are designed for a specific category, such as the Supplemental Nutrition Assistance Program (SNAP) and the Food Stamp Program in United States, in which case the voucher is distributed to the eligible households for food purchases (Hoynes and Schanzenbach 2009, Hastings and Shapiro 2018).

³ See Table 1 for the summary of the multiple coupon waves in Hangzhou.

⁴ A recent survey by PwC shows that the mobile payment penetration rate in China was 86% in 2019, the highest in the world, <https://www.pwc.com/gx/en/consumer-markets/consumer-insights-survey/2019/>

[report.pdf](#). Founded by the Alibaba group in 2004, Alipay is the leading mobile payment platform in China, with a market share of over 50%. Its market share is particularly high in Hangzhou, where Alibaba is headquartered. Our survey shows that 91.5% of the respondents use Alipay as their primary payment method.

⁵ The unique number of individuals is smaller than 1 million, because the same individuals could be in the control group for one coupon wave while in the treatment group for another wave.

⁶ We are not allowed to report summary statistics of some of these variables due to data sensitivity.

⁷ The statistics report that the total sales of catering and lodging businesses in China decreased by 35.3% in the first quarter of 2020 due to the COVID-19 pandemic. This sector alone absorbs 29.45 million employment in 2019.

⁸ The results are available upon request.

⁹ The results of the second-wave coupon yield qualitatively the same conclusions.

¹⁰ See online appendix for more details of the survey and the Guangxi coupon study.

¹¹ See Dellavigna (2009) for a survey on the literature of behavioral economics.

¹² The caveat is that the results may not extrapolate to the nonparticipating group. However, this is less of a concern, as the goal of this fiscal stimulus program is to target the most responsive individuals in the city of Hangzhou. Moreover, the results are based on large population, as about 2.3 million to 4.2 million individuals participated in each wave, which is 25% to 50% of the city population over the age of 18.

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